Mathematical Formulations and Algorithmic Analysis

of the Autonomous Mobile Robot (AMR) Project

Full Analytical Derivations: Kinematics, Dynamics, EKF, AMCL, Graph SLAM, MPPI, and Evasion Geometry

Autonomous Mobile Robot Engineering Team | System Architecture & Control Documentation

Compiled: October 29, 2026

Abstract

This document delivers an exhaustive mathematical and algorithmic breakdown of the Autonomous Mobile Robot (AMR) platform. The analysis covers every governing equation implemented in the project codebase, spanning differential-drive non-holonomic kinematics and 3D rigid-body inertia tensors, dual-source continuous-discrete Extended Kalman Filter (EKF) sensor fusion, Adaptive Monte Carlo Localization (AMCL) state sampling, Ceres-based pose-graph SLAM optimization, potential field costmap inflation, Euclidean Distance Transform (EDT) topological roadmap extraction, Model Predictive Path Integral (MPPI) stochastic control, C^1-continuous cubic Bézier reactive evasion trajectories, wavefront frontier exploration, pinhole camera projection geometry, and affine Canvas web dashboard coordinate transformations.

Contents

1 Robot Kinematics, Dynamics & Inertial Physics

The AMR is modeled as a two-wheel differential drive ground vehicle stabilized by front and rear passive low-friction spherical casters, operating in the planar special Euclidean group $SE(2)$.

1.1 Differential Drive Forward Kinematics

Let $(x, y, \theta)^T \in SE(2)$ denote the robot pose in the global inertial reference frame. The physical platform parameters derived from the unified robot description format (`warehouse_agv.urdf`) are:

- Wheel radius: $r = 0.025$ m (nominal wheel diameter 50 mm)
- Wheel track baseline: $L = 0.26$ m (wheel centers positioned at $y = \pm 0.13$ m)
- Chassis footprint radius: $R_{\text{base}} = 0.11$ m (chassis cylinder diameter 220 mm)
- Total mobile base mass: $M = 10.0$ kg

Given individual left and right wheel angular rotational velocities $\omega_L, \omega_R \in \mathbb{R}$, the forward translational velocity v and yaw rate ω in the robot body frame are defined by:

$$v = \frac{r}{2}(\omega_R + \omega_L) \tag{1}$$

$$\omega = \frac{r}{L}(\omega_R - \omega_L) \tag{2}$$

In matrix mapping form:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} \frac{r}{2} & \frac{r}{2} \\ \frac{r}{L} & -\frac{r}{L} \end{bmatrix} \begin{bmatrix} \omega_R \\ \omega_L \end{bmatrix} \tag{3}$$

The non-holonomic continuous-time state-space kinematic model in the global inertial frame is:

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos\theta(t) & 0 \\ \sin\theta(t) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v(t) \\ \omega(t) \end{bmatrix} \tag{4}$$

Subject to the non-holonomic zero lateral slip velocity constraint:

$$-\dot{x}(t)\sin\theta(t) + \dot{y}(t)\cos\theta(t) = 0 \tag{5}$$

1.2 Inverse Kinematics

Given an arbitrary commanded velocity twist $(v, \omega)^T$ issued by the trajectory controller, the required individual wheel angular velocities are calculated by inverting the kinematic matrix:

$$\omega_R = \frac{v + \frac{L}{2}\omega}{r} = \frac{2v + L\omega}{2r} \tag{6}$$

$$\omega_L = \frac{v - \frac{L}{2}\omega}{r} = \frac{2v - L\omega}{2r} \tag{7}$$

1.3 Exact Circular Arc Odometry Dead-Reckoning

Over discrete sampling period Δt, the left and right wheel displacement increments are $\Delta s_L = r\Delta\theta_L$ and $\Delta s_R = r\Delta\theta_R$. The net chord displacement Δs and rotational change $\Delta\theta$ are:

$$\Delta s = \frac{\Delta s_R + \Delta s_L}{2}, \quad \Delta\theta = \frac{\Delta s_R - \Delta s_L}{L} \tag{8}$$

When $\Delta\theta \neq 0$, the robot moves along an instantaneous circular arc of radius $R_{\text{arc}} = \frac{\Delta s}{\Delta\theta}$. The exact closed-form integral over $[t_k, t_{k+1}]$ is:

$$x_{k+1} = x_k + \Delta s \cdot \frac{\sin(\Delta\theta)}{\Delta\theta} \cos\left(\theta_k + \frac{\Delta\theta}{2}\right) \tag{9}$$

$$y_{k+1} = y_k + \Delta s \cdot \frac{\sin(\Delta\theta)}{\Delta\theta} \sin\left(\theta_k + \frac{\Delta\theta}{2}\right) \tag{10}$$

$$\theta_{k+1} = \theta_k + \Delta\theta \tag{11}$$

For quasi-straight motion where $|\Delta\theta| < 10^{-6}$, the third-order Taylor expansion replaces the division by zero:

$$\frac{\sin(\Delta\theta)}{\Delta\theta} \approx 1 - \frac{(\Delta\theta)^2}{6} + \mathcal{O}((\Delta\theta)^4) \tag{12}$$

$$x_{k+1} = x_k + \Delta s \cos\left(\theta_k + \frac{\Delta\theta}{2}\right), \quad y_{k+1} = y_k + \Delta s \sin\left(\theta_k + \frac{\Delta\theta}{2}\right) \tag{13}$$

1.4 Rigid-Body Mass & Moment of Inertia Tensors

The AMR's dynamic response and Gazebo physics simulation rely on 3D spatial inertia tensors:

1. **Cylindrical Chassis Body:** Radius $R = 0.11$ m, height $H = 0.08$ m, total mass $m = 10.0$ kg:

$$I_{xx} = I_{yy} = \frac{1}{12}m(3R^2 + H^2) = \frac{10}{12}(3(0.11)^2 + (0.08)^2) = 0.03558 \text{ kg} \cdot \text{m}^2 \tag{14}$$

$$I_{zz} = \frac{1}{2}mR^2 = \frac{1}{2}(10)(0.11)^2 = 0.06050 \text{ kg} \cdot \text{m}^2 \tag{15}$$

2. **Solid Drive Wheels:** Cylinders with mass $m_w = 0.5$ kg, radius $r = 0.025$ m, width $w = 0.02$ m:

$$I_{zz} = \frac{1}{2}m_w r^2 = \frac{1}{2}(0.5)(0.025)^2 = 1.5625 \times 10^{-4} \text{ kg} \cdot \text{m}^2 \tag{16}$$

$$I_{xx} = I_{yy} = \frac{1}{12}m_w(3r^2 + w^2) = \frac{0.5}{12}(3(0.025)^2 + (0.02)^2) = 9.479 \times 10^{-5} \text{ kg} \cdot \text{m}^2 \tag{17}$$

2 State Estimation & Sensor Fusion

2.1 Extended Kalman Filter (EKF Fusion)

The state estimation pipeline combines high-frequency wheel encoder dead-reckoning (50 Hz) and IMU angular rate / linear acceleration (50 Hz) using a 15-state non-linear continuous-discrete Extended Kalman Filter (`robot_localization`).

The full continuous state vector is:

$$\mathbf{x} = [x, y, z, \phi, \theta, \psi, \dot{x}, \dot{y}, \dot{z}, \dot{\phi}, \dot{\theta}, \dot{\psi}, \ddot{x}, \ddot{y}, \ddot{z}]^T \in \mathbb{R}^{15} \tag{18}$$

For planar ground navigation (`two_d_mode: true`), the operational state vector contracts to:

$$\mathbf{x}_{2D} = [x, y, \psi, \dot{x}, \dot{y}, \dot{\psi}, \ddot{x}, \ddot{y}]^T \in \mathbb{R}^8 \tag{19}$$

2.1.1 Continuous-Discrete Time State Prediction

The continuous-time process evolution model with zero-mean Gaussian white process noise $\mathbf{w}(t) \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_c)$ is:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t)) + \mathbf{w}(t) \tag{20}$$

For a sample period Δt, the discrete-time forward state propagation is:

$$\hat{\mathbf{x}}_{k|k-1} = \hat{\mathbf{x}}_{k-1|k-1} + \int_{t_{k-1}}^{t_k} \mathbf{f}(\hat{\mathbf{x}}(\tau))d\tau \tag{21}$$

The state transition Jacobian $\mathbf{F}_k = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\big|_{\hat{\mathbf{x}}_{k-1|k-1}}$ yields the discrete covariance prediction:

$$\mathbf{P}_{k|k-1} = \mathbf{\Phi}_k \mathbf{P}_{k-1|k-1} \mathbf{\Phi}_k^T + \mathbf{Q}_k, \quad \mathbf{\Phi}_k \approx \mathbf{I} + \mathbf{F}_k \Delta t \tag{22}$$

2.1.2 Discrete Measurement Updates

When an observation $\mathbf{z}_k \in \mathbb{R}^m$ arrives from wheel odometry or IMU with noise covariance $\mathbf{R}_k$:

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}) \quad \text{(Innovation Residual)} \tag{23}$$

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k \quad \text{(Innovation Covariance)} \tag{24}$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T \mathbf{S}_k^{-1} \quad \text{(Optimal Kalman Gain)} \tag{25}$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \mathbf{y}_k \quad \text{(State Update)} \tag{26}$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)^T + \mathbf{K}_k \mathbf{R}_k \mathbf{K}_k^T \quad \text{(Joseph Form Covariance)} \tag{27}$$

2.2 Adaptive Monte Carlo Localization (AMCL)

AMCL estimates the robot global pose $S_t = (x, y, \theta)^T$ using a particle filter $\{S_t^{[m]}, w_t^{[m]}\}_{m=1}^M$ conditioned on laser range scans z_t and map m.

2.2.1 Sampling Odometry Motion Model

Given dead-reckoning odometry displacement $\bar{x}_{t-1} = (\bar{x}, \bar{y}, \bar{\theta})^T$ and $\bar{x}_t = (\bar{x}', \bar{y}', \bar{\theta}')^T$, the relative motion parameters are:

$$\delta_{\text{rot1}} = \text{atan2}(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta} \tag{28}$$

$$\delta_{\text{trans}} = \sqrt{(\bar{x}' - \bar{x})^2 + (\bar{y}' - \bar{y})^2} \tag{29}$$

$$\delta_{\text{rot2}} = \bar{\theta}' - \bar{\theta} - \delta_{\text{rot1}} \tag{30}$$

Noisy perturbations are sampled using error parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4$:

$$\hat{\delta}_{\text{rot1}} = \delta_{\text{rot1}} - \text{sample}(\alpha_1 \delta_{\text{rot1}}^2 + \alpha_2 \delta_{\text{trans}}^2) \tag{31}$$

$$\hat{\delta}_{\text{trans}} = \delta_{\text{trans}} - \text{sample}(\alpha_3 \delta_{\text{trans}}^2 + \alpha_4 (\delta_{\text{rot1}}^2 + \delta_{\text{rot2}}^2)) \tag{32}$$

$$\hat{\delta}_{\text{rot2}} = \delta_{\text{rot2}} - \text{sample}(\alpha_1 \delta_{\text{rot2}}^2 + \alpha_2 \delta_{\text{trans}}^2) \tag{33}$$

2.2.2 Likelihood Field Sensor Model

Each LiDAR beam endpoint in global coordinates:

$$x_{z_t^k} = x + x_{\text{laser}} \cos\theta - y_{\text{laser}} \sin\theta + z_t^k \cos(\theta + \theta_{\text{laser}} + \phi_k) \tag{34}$$

$$y_{z_t^k} = y + x_{\text{laser}} \sin\theta + y_{\text{laser}} \cos\theta + z_t^k \sin(\theta + \theta_{\text{laser}} + \phi_k) \tag{35}$$

The likelihood probability mixes zero-delay Gaussian hit probability and random noise:

$$p(z_t^k | S_t, m) = z_{\text{hit}} \cdot \frac{1}{\sqrt{2\pi\sigma_{\text{hit}}^2}} \exp\left(-\frac{\text{dist}(x_{z_t^k}, y_{z_t^k}, m)^2}{2\sigma_{\text{hit}}^2}\right) + \frac{z_{\text{rand}}}{z_{\max}} \tag{36}$$

2.2.3 Kullback-Leibler Divergence (KLD) Particle Bounds

The dynamic particle count M required to keep the approximation error bounded by $\epsilon = 0.05$ with probability $1 - \delta = 0.99$ across k occupied bins is:

$$M \approx \frac{k-1}{2\epsilon} \left(1 - \frac{2}{9(k-1)} + \sqrt{\frac{2}{9(k-1)}} z_{1-\delta} \right)^3 \tag{37}$$

Clamped to the configured operational range: $M_{\min} = 500 \leq M \leq M_{\max} = 2000$.

3 SLAM & Pose-Graph Optimization

`slam_toolbox` optimizes robot trajectory poses $\mathbf{x} = \{x_1, \ldots, x_N\}$ and laser scan matches using non-linear least squares.

3.1 Correlative Scan Matching

Given current scan points $P = \{p_i\}$ and previous local map $\mathcal{M}$, the optimal rigid transform $\hat{\xi} = (\hat{t}_x, \hat{t}_y, \hat{\theta})^T \in SE(2)$ maximizes the correlation score:

$$\hat{\xi} = \arg\max_{\xi} \sum_{i=1}^{|P|} \mathcal{M}(T(\xi) \cdot p_i) \tag{38}$$

3.2 Ceres Pose-Graph Non-Linear Least Squares

The global trajectory cost functional over odometry edges $\mathcal{E}_{\text{odom}}$ and loop-closure constraints $\mathcal{E}_{\text{loop}}$ is:

$$F(\mathbf{x}) = \sum_{(i,j) \in \mathcal{E}} \mathbf{e}_{ij}(\mathbf{x}_i, \mathbf{x}_j)^T \mathbf{\Omega}_{ij} \mathbf{e}_{ij}(\mathbf{x}_i, \mathbf{x}_j) \tag{39}$$

Where the $SE(2)$ relative transformation error is:

$$\mathbf{e}_{ij}(\mathbf{x}_i, \mathbf{x}_j) = \begin{bmatrix} \mathbf{R}_i^T(\mathbf{t}_j - \mathbf{t}_i) - \mathbf{t}_{ij}^{\text{meas}} \\ \text{wrap2pi}(\theta_j - \theta_i - \Delta\theta_{ij}^{\text{meas}}) \end{bmatrix} \tag{40}$$

The non-linear system is iteratively solved using the Levenberg-Marquardt trust-region algorithm:

$$\left(\mathbf{J}^T \mathbf{\Omega} \mathbf{J} + \lambda \, \text{diag}(\mathbf{J}^T \mathbf{\Omega} \mathbf{J}) \right) \Delta \mathbf{x} = -\mathbf{J}^T \mathbf{\Omega} \mathbf{e} \tag{41}$$

4 Costmap Inflation & Potential Fields

The 2D Nav2 Costmap transforms binary obstacle geometry into continuous navigation risks.

$$C(d) = \begin{cases} 254 & \text{if } d = 0 \text{ (Lethal Obstacle)} \\ 253 & \text{if } 0 < d \leq R_{\text{inscribed}} \text{ (Inscribed Footprint Danger)} \\ \lfloor 252 \cdot \exp(-k \cdot (d - R_{\text{inscribed}})) \rfloor & \text{if } R_{\text{inscribed}} < d \leq R_{\text{inflation}} \\ 0 & \text{if } d > R_{\text{inflation}} \text{ (Free Navigable Space)} \end{cases} \tag{42}$$

Platform parameters:

- $R_{\text{inscribed}} = 0.13$ m (enclosing footprint radius $R_{\text{base}} = 0.11$ m plus 20 mm margin)
- $R_{\text{inflation}} = 0.25$ m (smooth repulsion buffer)
- Inflation decay scale: $k = 3.0$ m^{-1}

5 Topological Roadmap & Graph Theory

5.1 Euclidean Distance Transform (EDT)

Given binary grid occupancy $M(u, v) \in \{0, 1\}$, the distance to the nearest obstacle boundary is computed via the continuous transform:

$$D(u, v) = \min_{(u', v') : M(u', v') = 1} \sqrt{(u - u')^2 + (v - v')^2} \cdot \text{resolution} \tag{43}$$

5.2 Swept-Circle Clearance Verification

For an edge connecting topological nodes $\mathbf{p}_A$ and $\mathbf{p}_B$, collision-free safety is guaranteed by sampling along the parameterized chord:

$$\mathbf{p}(s) = (1-s)\mathbf{p}_A + s\mathbf{p}_B, \quad s \in [0,1] \tag{44}$$

$$\min_{s\in[0,1]} D(\mathbf{p}(s)) \geq R_{\text{robot_margin}} = 0.18 \text{ m} \tag{45}$$

5.3 Dijkstra Shortest Path Routing

The global planner extracts paths over graph $G = (V, E)$ by minimizing edge traversal distances:

$$w(u,v) = \|\mathbf{p}_u - \mathbf{p}_v\|_2 = \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2} \tag{46}$$

$$\text{dist}[v] = \min_{(u,v)\in E} (\text{dist}[u] + w(u,v)) \tag{47}$$

6 Local Motion Planning: Model Predictive Path Integral (MPPI)

The AMR uses an Model Predictive Path Integral (MPPI) controller for real-time trajectory optimization under dynamic obstacles.

6.1 Stochastic Trajectory Rollouts

At control cycle t_k, $K = 80$ stochastic candidate trajectories are sampled over a prediction horizon $T = 15$ steps ($\Delta t = 0.10$ s, total lookahead horizon 1.5 s):

$$v_k^{(i)} = \bar{v}_k + \epsilon_{v,k}^{(i)}, \quad \epsilon_v \sim \mathcal{N}(0, \sigma_v^2), \quad \sigma_v = 0.08 \text{ m/s} \tag{48}$$

$$\omega_k^{(i)} = \bar{\omega}_k + \epsilon_{\omega,k}^{(i)}, \quad \epsilon_\omega \sim \mathcal{N}(0, \sigma_\omega^2), \quad \sigma_\omega = 0.35 \text{ rad/s} \tag{49}$$

Each trajectory is propagated forward using non-holonomic unicycle integration:

$$x_{t+1}^{(i)} = x_t^{(i)} + v_t^{(i)} \cos(\theta_t^{(i)}) \Delta t \tag{50}$$

$$y_{t+1}^{(i)} = y_t^{(i)} + v_t^{(i)} \sin(\theta_t^{(i)}) \Delta t \tag{51}$$

$$\theta_{t+1}^{(i)} = \theta_t^{(i)} + \omega_t^{(i)} \Delta t \tag{52}$$

6.2 Composite Cost Functional

The total cost $S(\tau^{(i)})$ of rollout $\tau^{(i)}$ balances six objective criteria:

$$S(\tau^{(i)}) = J_{\text{term}} + J_{\text{heading}} + J_{\text{cte}} + J_{\text{repulsion}} + J_{\text{dyn_coll}} + J_{\text{effort}} \tag{53}$$

1. **Terminal Waypoint Distance Cost:**

$$J_{\text{term}} = w_{\text{dist}} \cdot \left\| \mathbf{p}_T^{(i)} - \mathbf{p}_{\text{target}} \right\|_2, \quad w_{\text{dist}} = 4.0 \tag{54}$$

2. **Horizon Heading Alignment Cost:**

$$J_{\text{heading}} = w_{\text{heading}} \cdot \left| \text{wrap2pi}\left(\theta_T^{(i)} - \text{atan2}(y_{\text{tgt}} - y_T^{(i)}, x_{\text{tgt}} - x_T^{(i)}) \right) \right|, \quad w_{\text{heading}} = 3.0 \tag{55}$$

3. **Cross-Track Error (CTE) Cost:**

$$J_{\text{cte}} = w_{\text{cte}} \sum_{t=1}^{T} d_\perp(\mathbf{p}_t^{(i)}, \text{Path Segment}), \quad w_{\text{cte}} = 6.0 \text{ (nominal)}, \quad 2.0 \text{ (evasion)} \tag{56}$$

4. **Static Obstacle Distance Repulsion:**

$$J_{\text{repulsion}} = w_{\text{rep}} \sum_{t=1}^{T} \max\left(0, d_{\text{safe}} - \text{dist}(\mathbf{p}_t^{(i)}, \text{Obstacles})\right)^2 \tag{57}$$

5. **Predictive Dynamic Obstacle Collision Penalty:**

$$J_{\text{dyn_coll}} = \sum_{t=1}^{T} \sum_{m=1}^{M_{\text{obs}}} \begin{cases} w_{\text{coll}} & \text{if } \|\mathbf{p}_t^{(i)} - \hat{\mathbf{p}}_{m,t}\|_2 < R_{\text{coll}} = 0.28 \text{ m} \\ 0 & \text{otherwise} \end{cases} \tag{58}$$

where $w_{\text{coll}} = 5000.0$.

6.3 Softmax Optimal Action Selection

The optimal command vector $\mathbf{u}_t^* = (v_t^*, \omega_t^*)^T$ is evaluated as the importance-weighted expectation:

$$\beta = \min_{i=1,\dots,K} S(\tau^{(i)}) \tag{59}$$

$$w^{(i)} = \frac{\exp\left(-\frac{1}{\lambda}(S(\tau^{(i)}) - \beta)\right)}{\sum_{j=1}^{K} \exp\left(-\frac{1}{\lambda}(S(\tau^{(j)}) - \beta)\right)}, \quad \lambda = 0.50 \tag{60}$$

$$\mathbf{u}_t^* = \sum_{i=1}^{K} w^{(i)} \mathbf{u}_t^{(i)} \tag{61}$$

6.4 Slew-Rate Limiting & EMA Velocity Smoothing

To ensure actuator continuity and eliminate jerking motion:

$$v_{\text{slew}} = \begin{cases} \min(v_t^*, v_{t-1} + a_{\text{accel}}\Delta t) & \text{if } v_t^* \geq v_{t-1} \\ \max(v_t^*, v_{t-1} - a_{\text{decel}}\Delta t) & \text{if } v_t^* < v_{t-1} \end{cases} \tag{62}$$

$$\omega_{\text{slew}} = \text{clamp}\left(\omega_t^*, \omega_{t-1} - \alpha_{\text{accel}}\Delta t, \omega_{t-1} + \alpha_{\text{accel}}\Delta t\right) \tag{63}$$

Where $a_{\text{accel}} = 0.60$ m/s^2, $a_{\text{decel}} = 0.90$ m/s^2, and $\alpha_{\text{accel}} = 1.60$ rad/s^2.

Finally, an Exponential Moving Average (EMA) filter is applied:

$$v_{\text{final}} = \alpha_v \cdot v_{\text{slew}} + (1 - \alpha_v) \cdot v_{\text{prev}}, \quad \alpha_v = 0.70 \tag{64}$$

$$\omega_{\text{final}} = \alpha_\omega \cdot \omega_{\text{slew}} + (1 - \alpha_\omega) \cdot \omega_{\text{prev}}, \quad \alpha_\omega = 0.70 \tag{65}$$

7 Dynamic Obstacle Tracking & Evasion Geometry

7.1 Clustering and Velocity Estimation

LiDAR scan endpoints are clustered using Euclidean radius filtering ($d_{\text{cluster}} = 0.35$ m). The cluster centroid is:

$$\bar{\mathbf{p}}_k = \frac{1}{|C|} \sum_{p_i \in C} p_i \tag{66}$$

Obstacle velocity is estimated by finite differences over timestamp separation Δt:

$$\mathbf{v}_{\text{obs}} = \frac{\bar{\mathbf{p}}_k - \bar{\mathbf{p}}_{k-1}}{\Delta t} \tag{67}$$

7.2 Vector Cross-Product Swerve Decision

To commit to a consistent evasion corridor (left versus right avoidance):

$$\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{robot}} - \mathbf{v}_{\text{obs}} \tag{68}$$

$$\mathbf{r}_{\text{obs}} = \mathbf{p}_{\text{obs}} - \mathbf{p}_{\text{robot}} \tag{69}$$

The 2D scalar cross-product defines the evasion commitment sign:

$$\zeta = \mathbf{v}_{\text{rel}} \times \mathbf{r}_{\text{obs}} = v_{\text{rel},x} r_{\text{obs},y} - v_{\text{rel},y} r_{\text{obs},x} \tag{70}$$

$$\text{Side} = \begin{cases} +1 \text{ (Left Swerve)} & \text{if } \zeta > 0 \\ -1 \text{ (Right Swerve)} & \text{if } \zeta \leq 0 \end{cases} \tag{71}$$

7.3 C^1-Continuous Cubic Bézier Detour Trajectory

The evasion path is synthesized as a parametric cubic polynomial curve:

$$\mathbf{B}(u) = (1-u)^3\mathbf{P}_0 + 3(1-u)^2u\mathbf{P}_1 + 3(1-u)u^2\mathbf{P}_2 + u^3\mathbf{P}_3, \quad u \in [0,1] \tag{72}$$

With control point anchors:

- $\mathbf{P}_0 = \mathbf{p}_{\text{start}}$
- $\mathbf{P}_1 = \mathbf{P}_0 + \kappa \cdot \mathbf{t}_{\text{start}}$ (satisfying velocity alignment $\mathbf{B}'(0) = 3(\mathbf{P}_1 - \mathbf{P}_0)$)
- $\mathbf{P}_2 = \mathbf{P}_3 - \kappa \cdot \mathbf{t}_{\text{end}}$
- $\mathbf{P}_3 = \mathbf{p}_{\text{rejoin}}$

8 Autonomous Exploration & Multi-Robot Map Merging

8.1 Wavefront Frontier Detection (WFD)

A frontier point $\mathbf{x}_f$ satisfies:

$$M(\mathbf{x}_f) = \text{Free} \quad \wedge \quad \exists \mathbf{x}_{\text{nbr}} \in \mathcal{N}_8(\mathbf{x}_f) \text{ s.t. } M(\mathbf{x}_{\text{nbr}}) = \text{Unknown} \tag{73}$$

8.2 Exploration Utility Function

Candidate frontier clusters are scored by trading off expected information gain against navigation cost:

$$U(f) = w_{\text{gain}} \cdot \text{Area}(f) - w_{\text{cost}} \cdot \text{Cost}_{\text{Dijkstra}}(\mathbf{p}_{\text{robot}}, \text{centroid}(f)) \tag{74}$$

8.3 Multi-Robot $SE(2)$ Map Merging

Given maps $\mathcal{M}_A$ and $\mathcal{M}_B$, the rigid transform $(\mathbf{R}, \mathbf{t})$ aligns features via RANSAC:

$$\mathbf{p}_i^A \approx \mathbf{R}\mathbf{p}_i^B + \mathbf{t} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_i^B \\ y_i^B \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \tag{75}$$

Merged cell values are fused in log-odds space:

$$L(m_{x,y}) = \ln\left(\frac{P(m_{x,y})}{1 - P(m_{x,y})}\right) \tag{76}$$

$$L_{\text{merged}} = L_A(x,y) + L_B(T(x,y)) - L_0 \tag{77}$$

9 Computer Vision Geometry

9.1 Pinhole Camera Intrinsic Projection Matrix

A 3D spatial landmark point $\mathbf{P}_c = (X_c, Y_c, Z_c)^T$ in camera coordinates projects to image plane pixel coordinates $(u, v)^T$:

$$s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}\mathbf{P}_c = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} \tag{78}$$

$$u = f_x\frac{X_c}{Z_c} + c_x, \quad v = f_y\frac{Y_c}{Z_c} + c_y \tag{79}$$

9.2 Normalized Pixel Offset & Depth Proxy

The horizontal offset angle ψ for detected objects is:

$$\Delta u = \frac{u_{\text{bbox_center}} - c_x}{W/2} \in [-1, 1] \tag{80}$$

$$\psi = \arctan\left(\frac{\Delta u \cdot (W/2)}{f_x}\right) \tag{81}$$

Distance is approximated using known physical height H_{obj}:

$$Z_{\text{est}} = \frac{f_y \cdot H_{\text{obj}}}{h_{\text{bbox}}} \tag{82}$$

10 Web Dashboard Coordinate Transformations

The HTML5 Canvas dashboard maps continuous ROS world coordinates (x_w, y_w) to display pixels (p_x, p_y):

$$p_x = \frac{x_w - x_0}{\text{resolution}} \tag{83}$$

$$p_y = H_{\text{map}} - \frac{y_w - y_0}{\text{resolution}} \tag{84}$$

With user zoom scale s and pan translation (T_x, T_y), screen coordinates (S_x, S_y) are:

$$\begin{bmatrix} S_x \\ S_y \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & T_x \\ 0 & s & T_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ 1 \end{bmatrix} \tag{85}$$

Smoothed robot orientation and position render via 60 FPS Linear Interpolation (LERP):

$$x(t) = (1-\gamma)x(t-1) + \gamma x_{\text{new}}, \quad \gamma = 0.25 \tag{86}$$

$$\theta(t) = \theta(t-1) + \gamma \cdot \operatorname{atan2}\left(\sin(\theta_{\text{new}} - \theta(t-1)), \cos(\theta_{\text{new}} - \theta(t-1))\right) \tag{87}$$

11 System Constants Reference Table

Parameter Name	**Symbol**	**Value & Units**	**Subsystem Reference**
Wheel Radius	r	0.025 m	URDF Kinematics
Wheel Baseline Track	L	0.26 m	URDF Kinematics
Chassis Mass	M	10.0 kg	Rigid Body Dynamics
Chassis Radius	R_{base}	0.11 m	Footprint Clearance
Inscribed Radius	$R_{\text{inscribed}}$	0.13 m	Nav2 Costmap
Inflation Radius	$R_{\text{inflation}}$	0.25 m	Nav2 Potential Field
Costmap Decay Rate	k	3.0 m^{-1}	Exponential Costmap
Rollout Trajectories	K	80	MPPI Controller
Prediction Steps	T	15	MPPI Controller
Time Step Horizon	Δt	0.10 s	MPPI Controller
Temperature Parameter	λ	0.50	MPPI Softmax
Linear Acceleration Limit	a_{accel}	0.60 m/s^2	Slew Rate Limiter
Linear Deceleration Limit	a_{decel}	0.90 m/s^2	Slew Rate Limiter
Angular Acceleration Limit	α_{accel}	1.60 rad/s^2	Slew Rate Limiter
Velocity EMA Filter Alpha	α_v, α_ω	0.70	Low-Pass Filter
Collision Risk Radius	R_{coll}	0.28 m	Dynamic Collision Avoidance
Min Particles	M_{min}	500	AMCL Particle Filter
Max Particles	M_{max}	2000	AMCL Particle Filter
Map Resolution	res	0.05 m/pixel	SLAM Occupancy Grid

End of Autonomous Mobile Robot (AMR) Mathematical Formulations Report.